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(54) **BOTTOM CAP FOR FOAM BUILDING PANELS**

(52) **U.S. Cl.**
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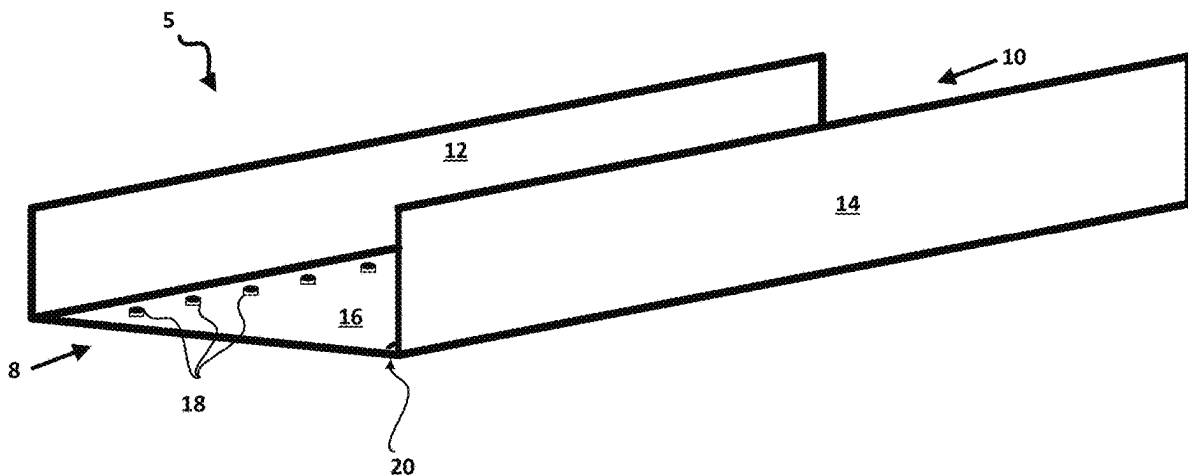
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(57) **ABSTRACT**

A bottom cap for a foam building panel includes an elongate rear wall; an elongate front wall spaced from and extending parallel to the elongate rear wall; a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel; a first set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate rear wall; and a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall. The structure of the bottom cap provides capping for the foam building panel but also some spacing and draining capabilities for enabling moisture that has entered between the bottom cap and the foam building panel to escape.



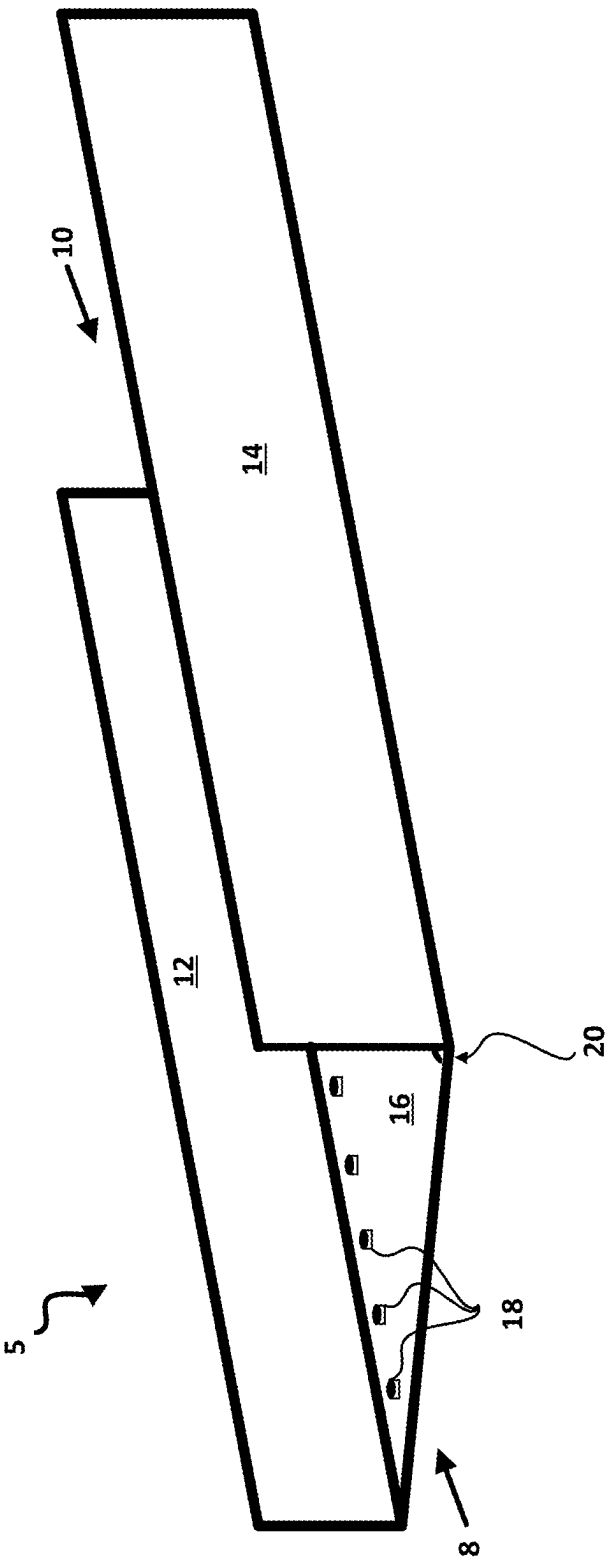


FIG. 1

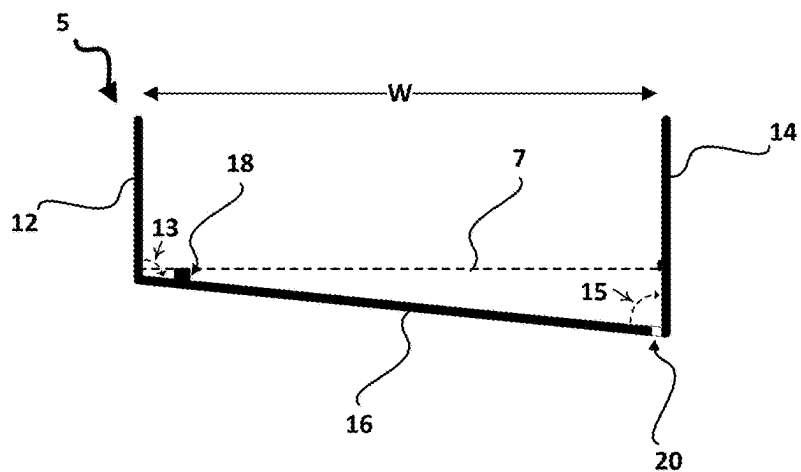


FIG. 2

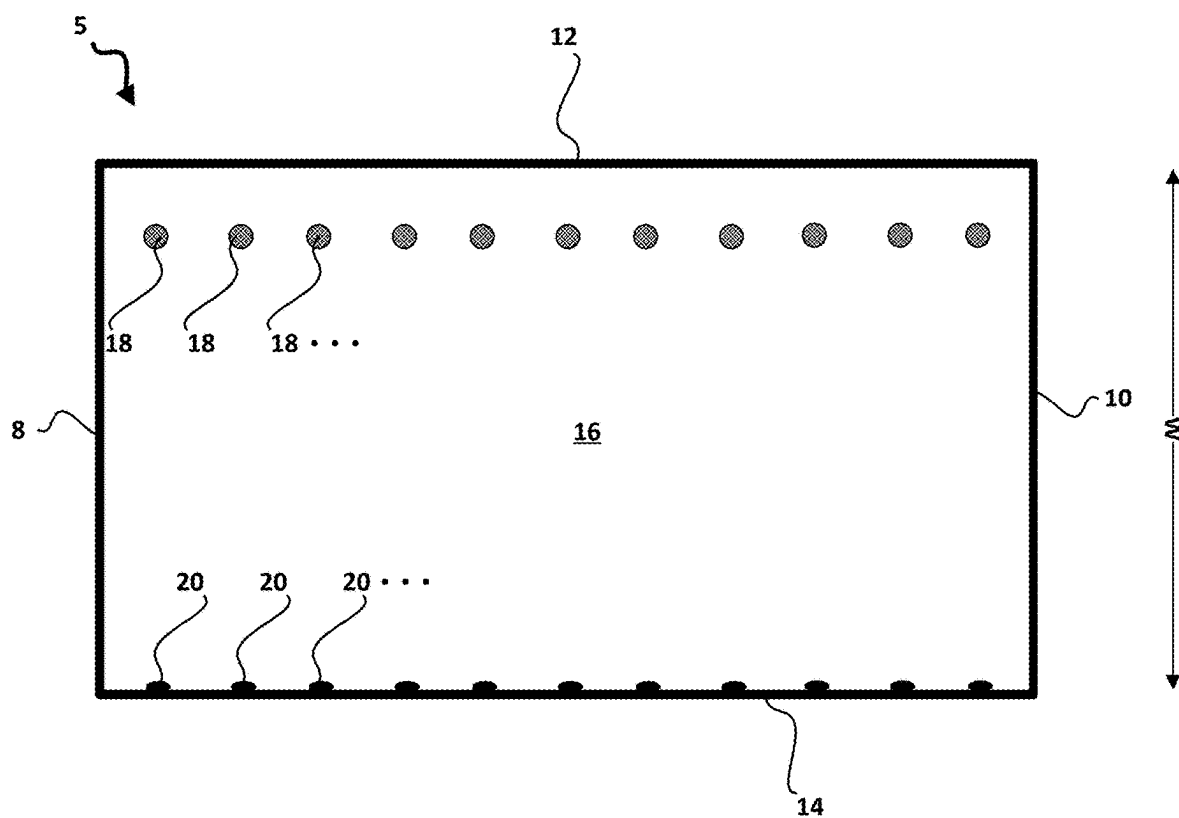
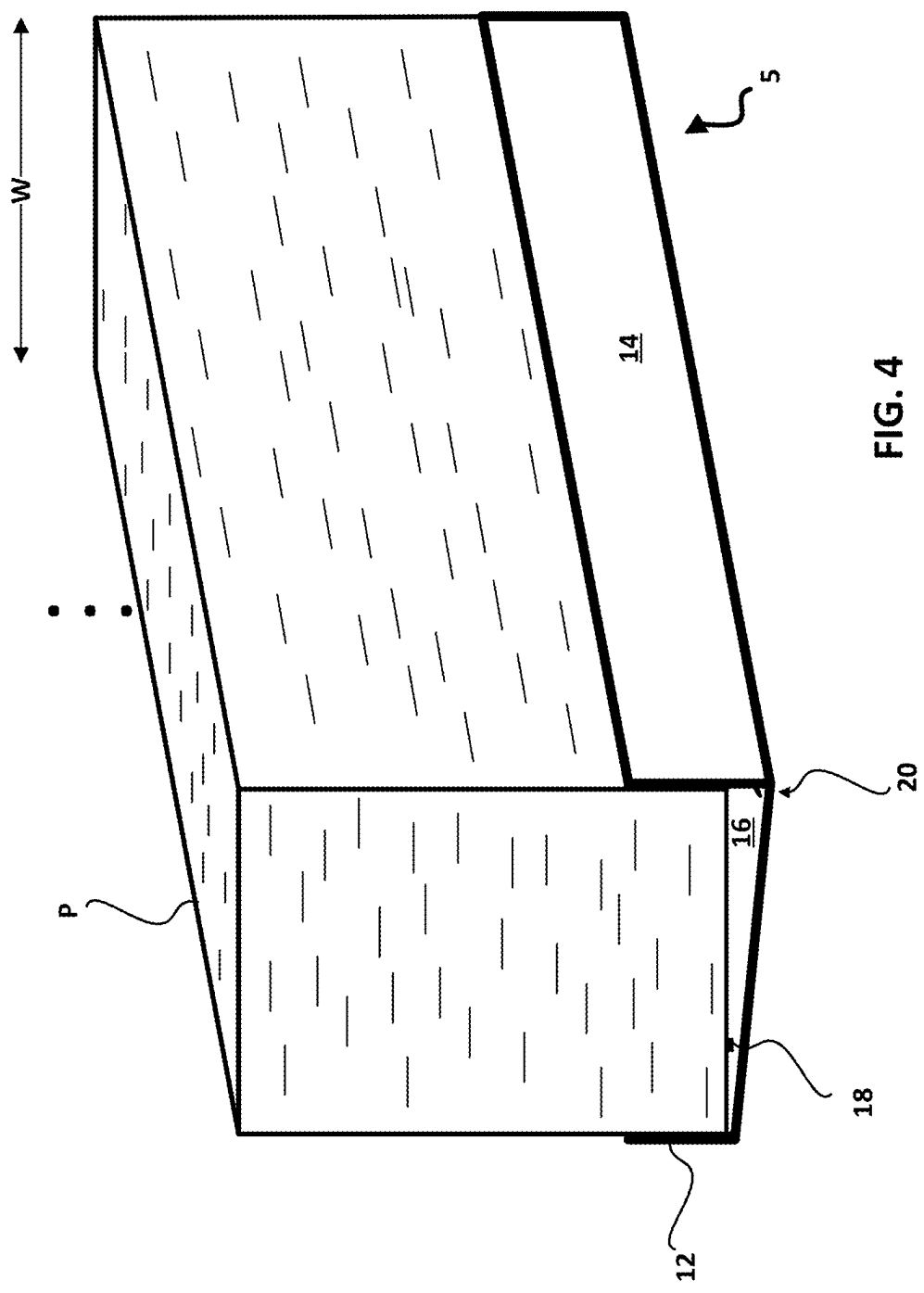
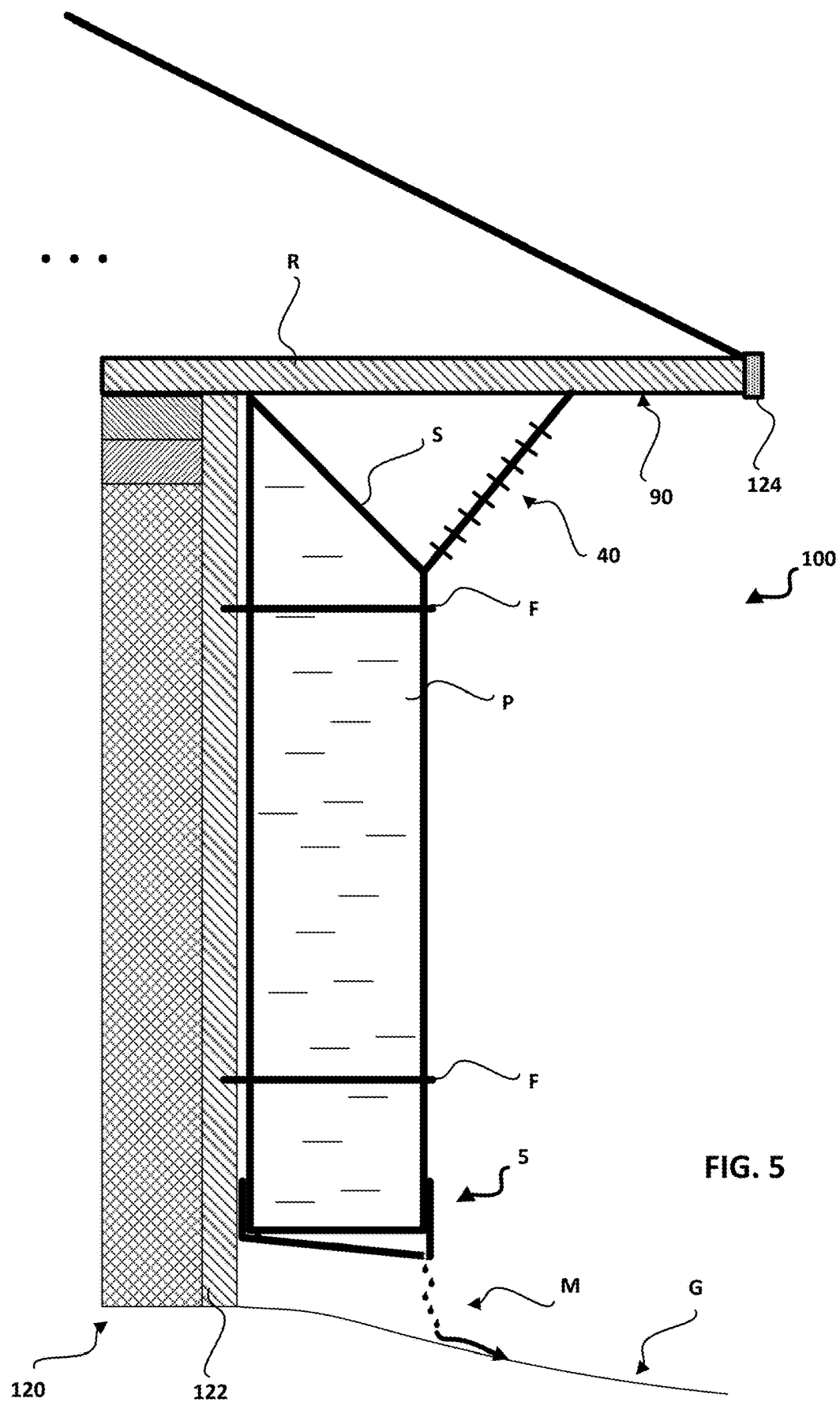


FIG. 3





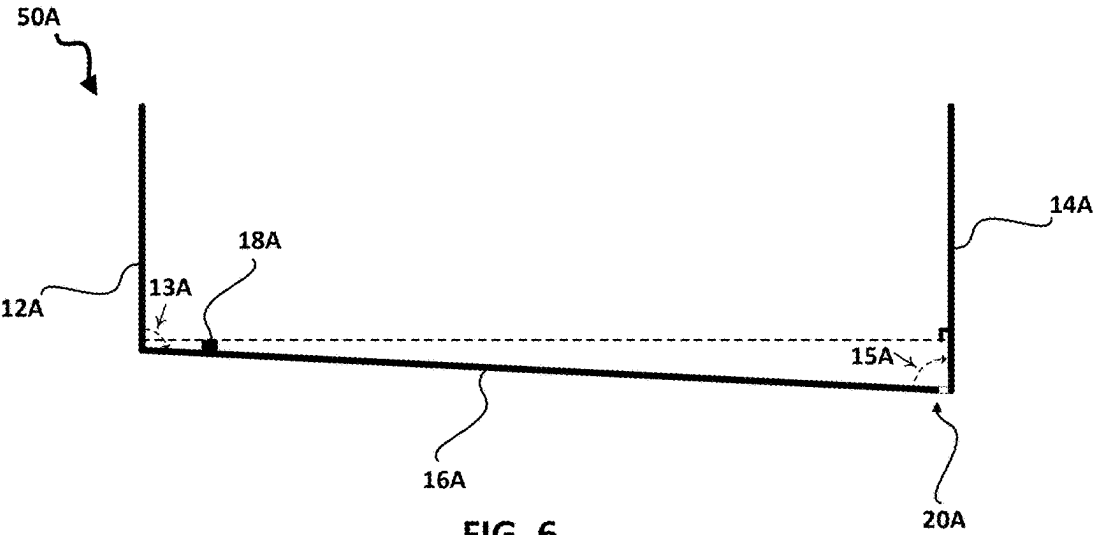


FIG. 6

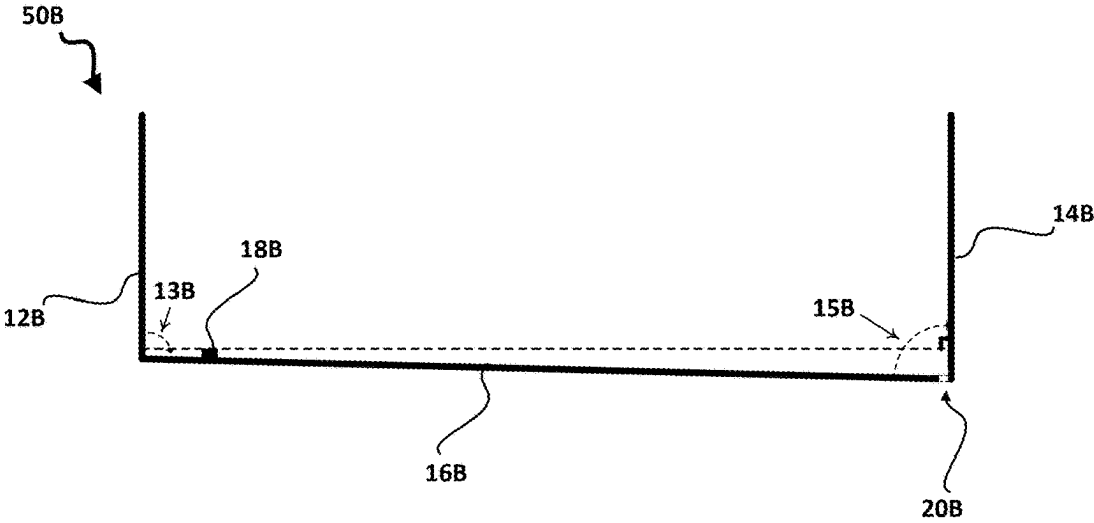


FIG. 7

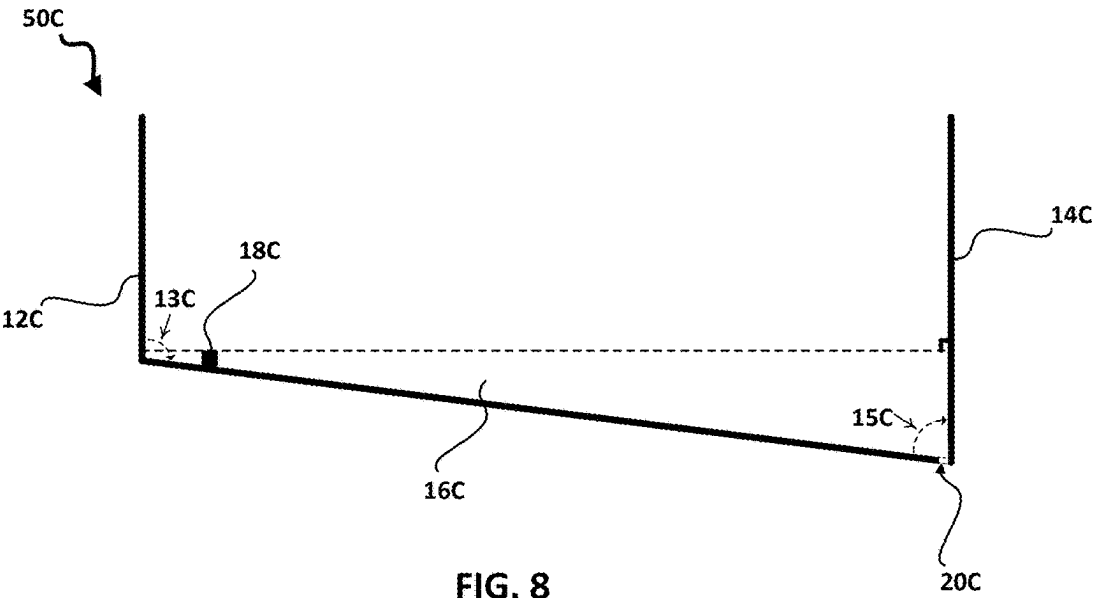
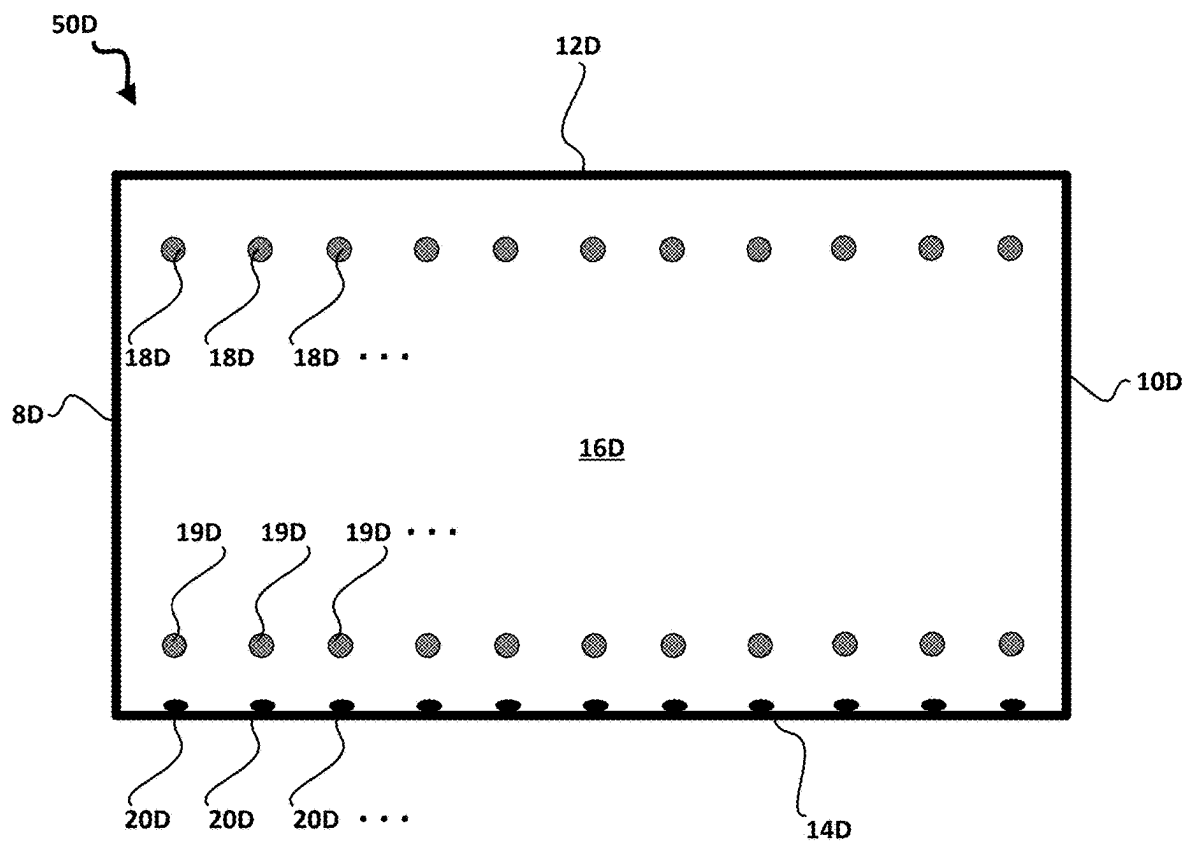
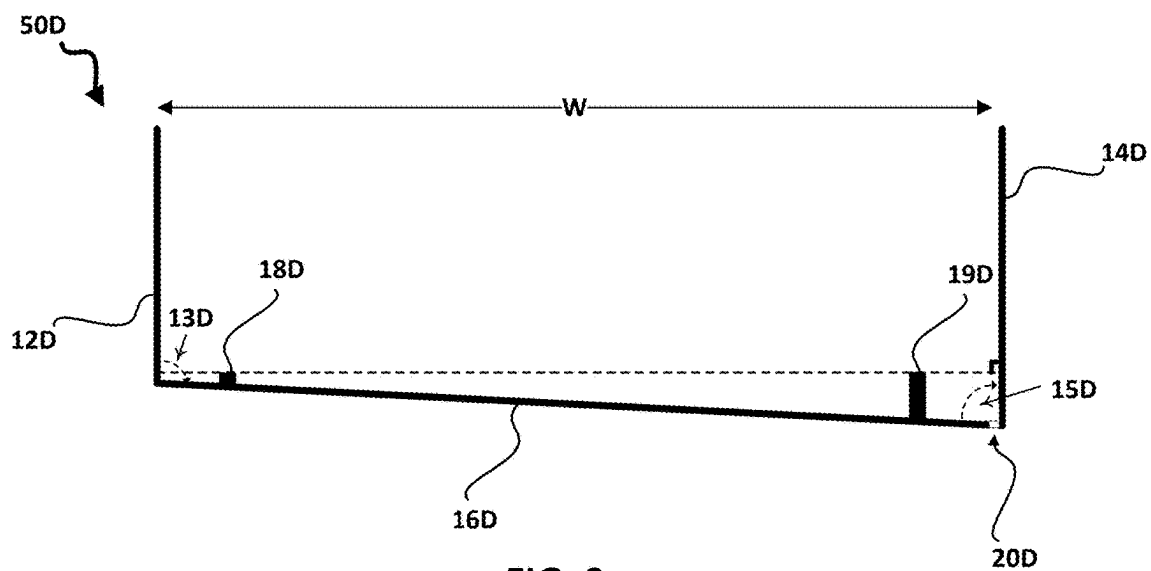
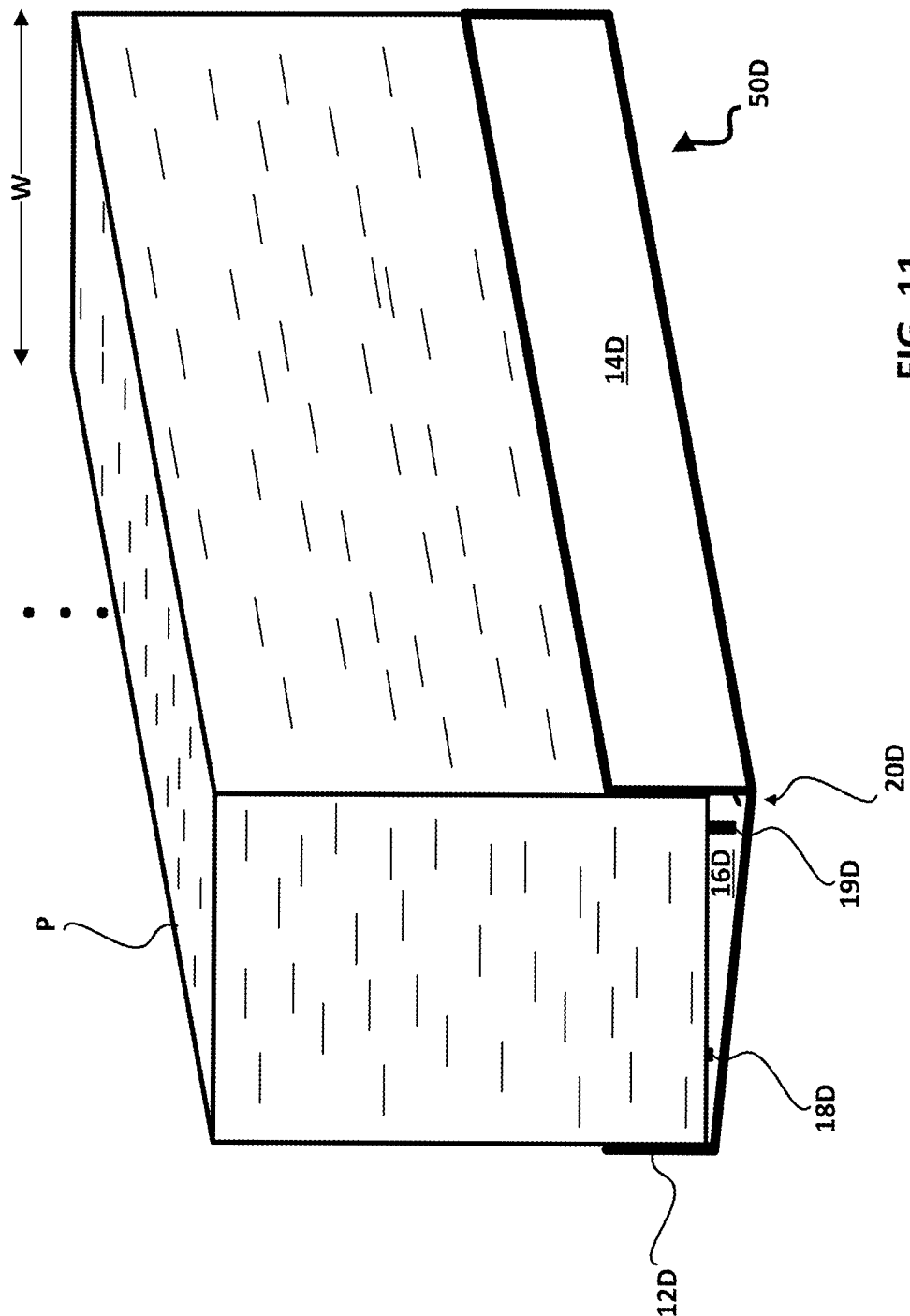


FIG. 8





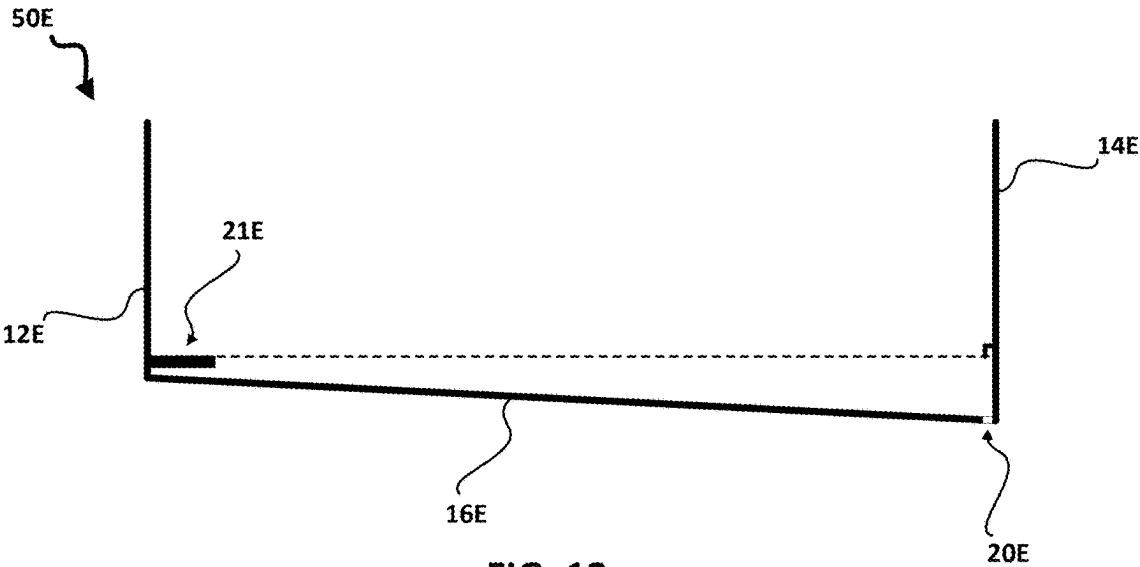


FIG. 12

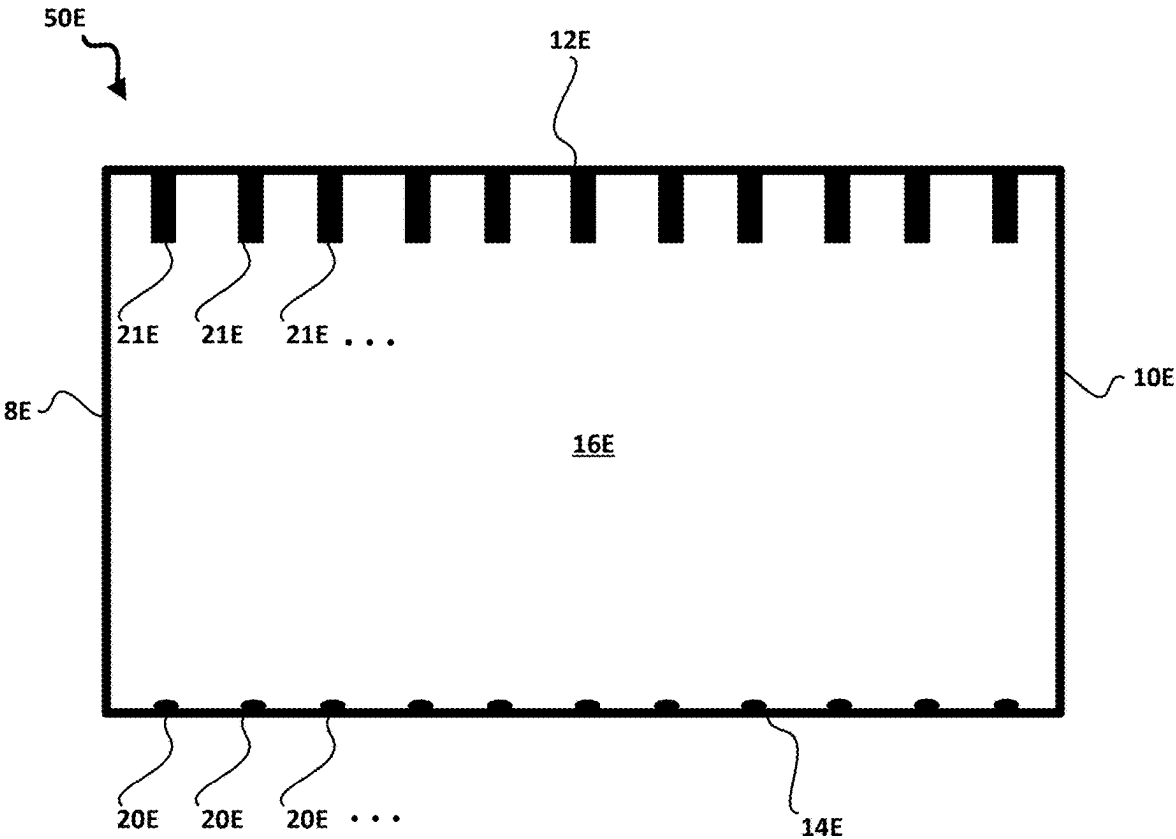


FIG. 13

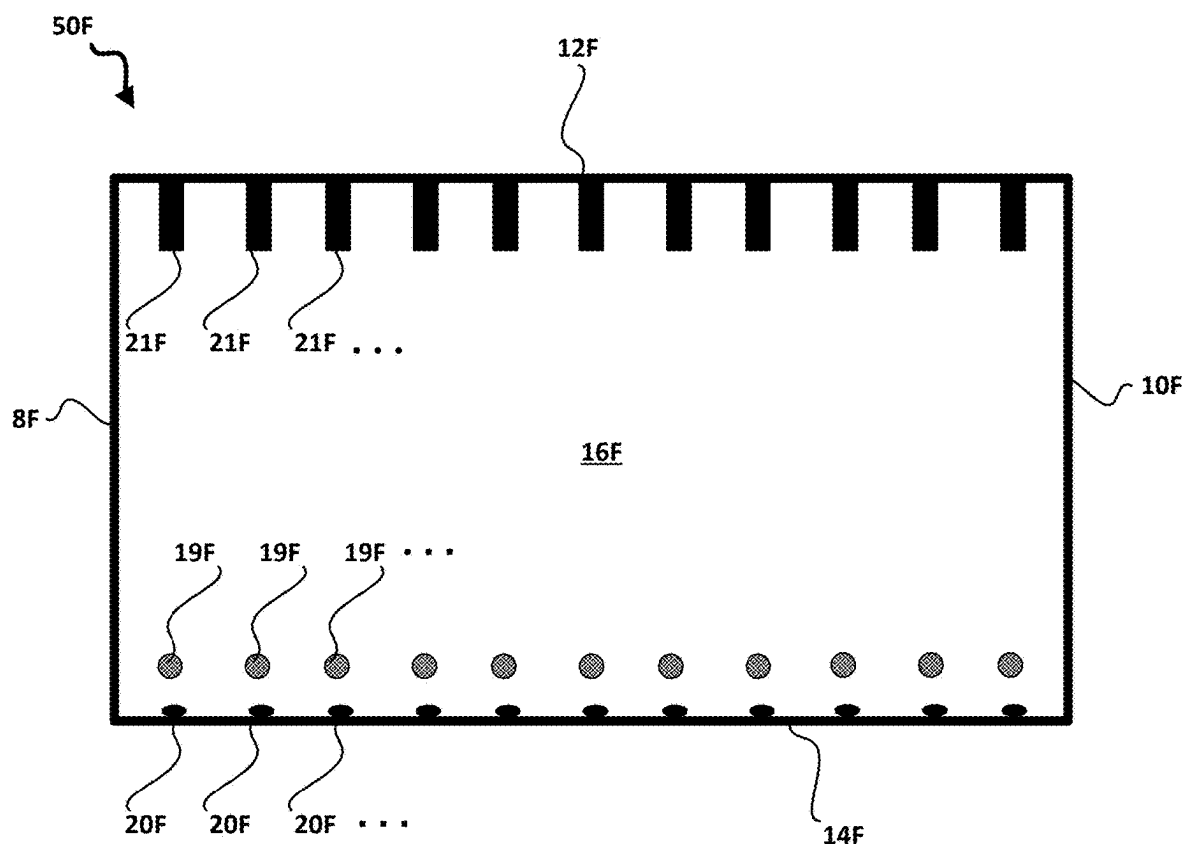
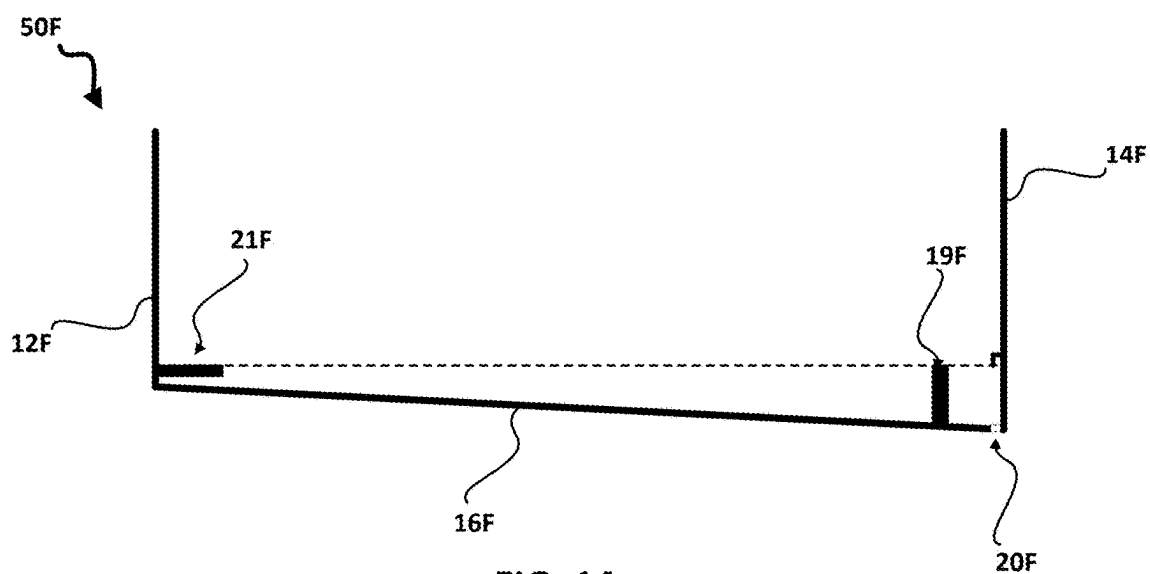
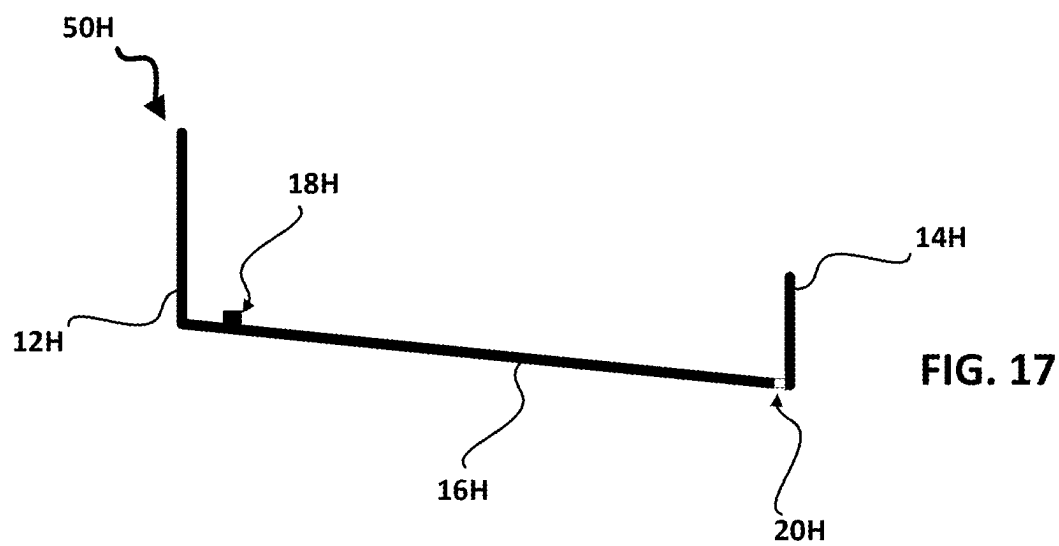
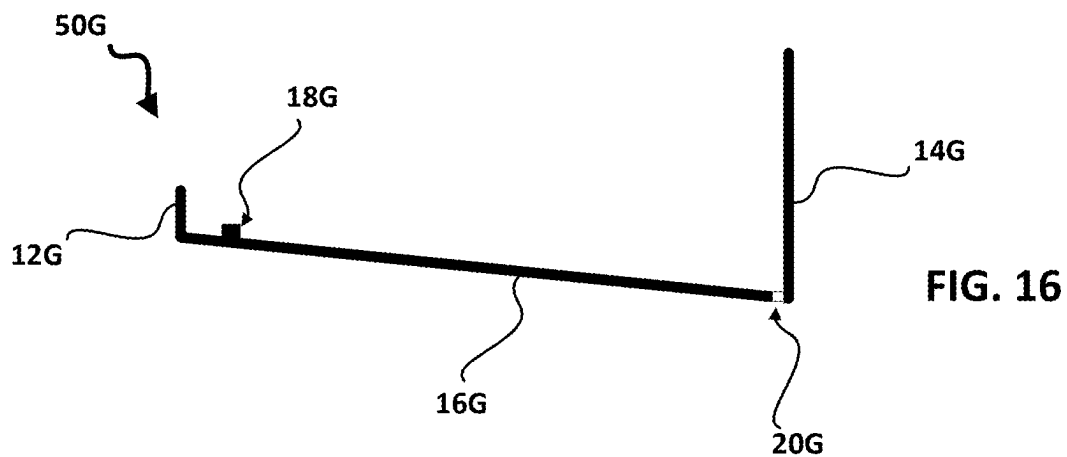
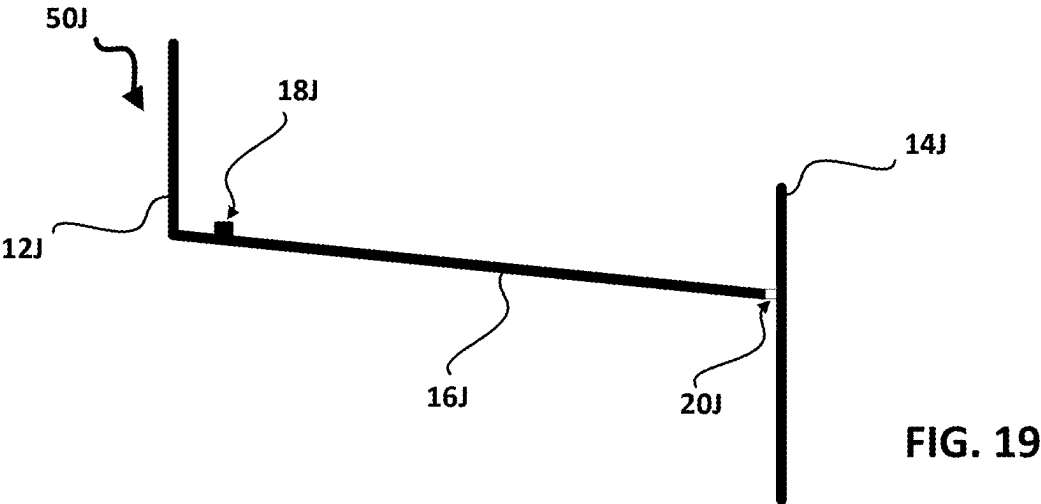
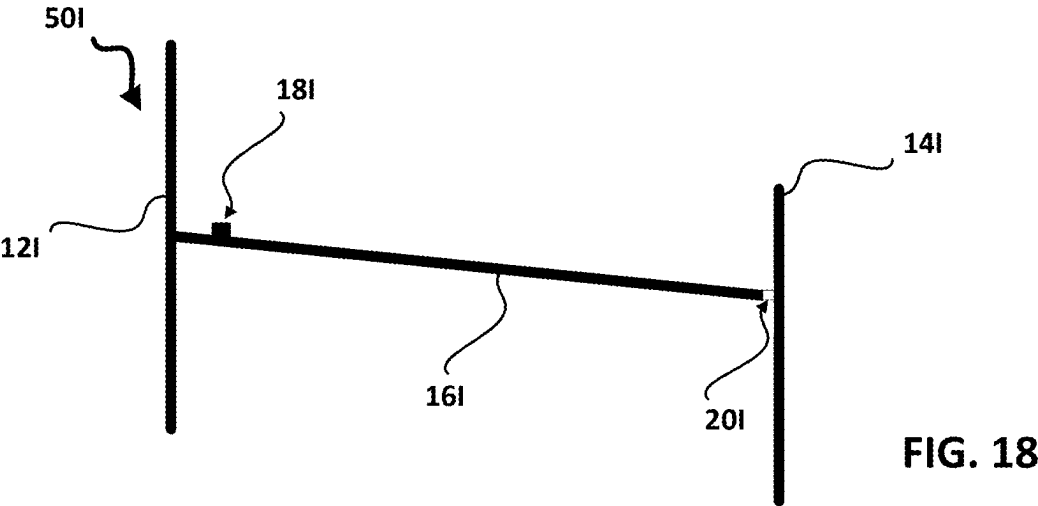


FIG. 15





BOTTOM CAP FOR FOAM BUILDING PANELS

CROSS REFERENCE

[0001] This application claims priority to U.S. Provisional Patent Application Ser. No. 63/563,345 filed on Mar. 9, 2024 entitled "BOTTOM CAP FOR FOAM BUILDING PANELS," the content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates generally to building systems, and more particular to a bottom cap for an exterior cladding panel system for buildings.

BACKGROUND

[0003] The construction of a building or dwelling is regulated by a set of standards, referred to collectively when contemplating construction, materials and the like as the building code. The building code is constantly evolving, and is intended to establish a minimum requirement for material dimensions and structural requirements with the objective of providing safe, comfortable and healthy environments.

[0004] Energy efficiency has received significant attention in the evolution of the building code, with a view to increasing energy efficiency or, at least, reducing energy consumption thereby to, in consequence, reduce the levels of CO₂ in the atmosphere and the costs of operating a building. Newer buildings have been increasingly energy efficient, due to being increasingly insulated. However, with the increases in energy efficiency due to insulation, it has also been recognized that careful attention must be paid to ensuring proper ventilation within the building envelope to avoid condensation.

[0005] An area of particular focus for ventilation is the roof space. In roof space, hot moist air from within the building is subject to cooler ambient temperatures, leading to excessive condensation. Such condensation can lead to undesirable deterioration of the building materials. To avoid this, the building code in many jurisdictions mandates ventilation of the roof space to inhibit such condensation.

[0006] One scheme for ensuring adequate ventilation involves providing the building with an overhanging soffit running around the periphery of the building's roof, there being portions of the soffit that are perforated or that otherwise permit fluid flow along with one or more ridge or gable vents at elevated locations to promote continuous air circulation in the roof space. The soffit is an integral part of the building structure and is typically incorporated into the design of the roof trusses and/or of rafters. In particular, the roof extends outward from the wall a sufficient distance to provide the ventilation area through the soffit required by the building code after the wall cladding and fascia has been applied.

[0007] With the ever-present desire to maximize profits by keeping material to a minimum, the overhang of the soffit is often the minimum allowed by the building code at the time the house was built. In such arrangements, the entire soffit may be perforated to meet the ventilation requirements of the building code.

[0008] As for insulation, with new buildings, insulation can be integrated into walls during construction to ensure the buildings comply with the latest insulation requirements in

the building code. However, for existing buildings other approaches are required to ensure the buildings better approach the insulation standards of the latest building code. For example, one approach is to add insulation to the exterior wall structure of an existing building. In particular, one may add sheet insulation to the exterior wall structure, using rectangular panels or sheets made of several inches of expanded polystyrene (EPS) such as Styrofoam™ available from Dow Chemical Corporation. After such insulating panels have been installed, a variety of finishes can be used to clad the exterior walls.

[0009] Adding EPS panels to the exterior of a building involves arranging the panels to abut the overhang. However, this results in several inches of the perforated soffit, corresponding to the several-inches thickness of the panels and any further cladding applied thereto, being blocked. This accordingly reduces free flow of air into and out of the building. As such, while the insulation has been improved, this consequent reduction in air flow can cause the building code to be contravened for failure to maintain adequate ventilation. Furthermore, it is generally not practical or cost effective to entirely restructure soffits to compensate for the blockages using existing techniques.

[0010] U.S. Pat. No. 10,301,817 to Ackerley, the contents of which are incorporated herein by reference, discloses an exterior insulating system for a building includes at least one insulating panel having a wall-facing side and an outward-facing side, a top and a bottom. The at least one insulating panel having a slope at the top that angles downwardly from the wall-facing side to the outward-facing side. The system includes at least one elongated clip member fastenable along a soffit that is extending outwards from a wall of the building, as well as at least one auxiliary panel having a first edge that is dimensioned to be received by the at least one elongated clip member opposite a main body of the at least one auxiliary panel from a second edge of the at least one auxiliary panel that is connectable to the at least one insulating panel. The main body incorporates structure for permitting airflow through the main body.

[0011] In the system disclosed in the '817 patent, a bottom of the insulating panel facing the foot of the building may be left exposed, though it may be useful to cover the bottom of the insulating panel for aesthetic and functional reasons, such as for preventing animals, insects and weather from reaching the bottom of the insulating panel and potentially breaking it into pieces over time. However, completely covering the bottom of an insulating panel may prevent any moisture that may arise behind the insulating panel from escaping the system through the bottom, causing a moisture buildup which can lead to problems with freezing and decay.

[0012] It is an object of the present disclosure to obviate or mitigate the above disadvantages.

SUMMARY

[0013] In accordance with an aspect, there is provided a bottom cap for a foam building panel, the bottom cap comprising: an elongate rear wall; an elongate front wall spaced from and extending parallel to the elongate rear wall; a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel; a first set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate rear wall; and a set of spaced apertures extending from inside the open-

topped channel through a region of the bottom wall that is proximal to the elongate front wall.

[0014] In an example, the obtuse inside angle is from about 95 degrees to about 105 degrees.

[0015] In an example, the obtuse inside angle is from about 97 degrees to about 102 degrees.

[0016] In an example, the obtuse inside angle is 97 degrees.

[0017] In an example, the obtuse inside angle is 102 degrees.

[0018] In an example, the bottom cap further comprises: a second set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the second set extends to a same height with respect to the elongate rear wall as the spaced pedestals in the first set.

[0019] In an example, the elongate front wall is spaced from the elongate rear wall by about 4 inches.

[0020] In an example, the elongate rear wall extends to about 1 inch from the bottom wall.

[0021] In an example, the elongate front wall extends to about 1.5 inches from the bottom wall.

[0022] In an example, the elongate front wall extends to about 1.85 inches from the bottom wall.

[0023] In accordance with another aspect, there is provided a bottom cap for a foam building panel, the bottom cap comprising: an elongate rear wall; an elongate front wall spaced from and extending parallel to the elongate rear wall; a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel; a set of spaced ledges extending into the open-topped channel from the elongate rear wall; and a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.

[0024] In an example, the obtuse inside angle is from about 95 degrees to about 105 degrees.

[0025] In an example, the obtuse inside angle is from about 97 degrees to about 102 degrees.

[0026] In an example, the obtuse inside angle is 97 degrees.

[0027] In an example, the obtuse inside angle is 102 degrees.

[0028] In an example, the bottom cap further comprises: a set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the second set extends to a same height with respect to the elongate rear wall as the spaced ledges.

[0029] In an example, the elongate front wall is spaced from the elongate rear wall by about 4 inches.

[0030] In an example, the elongate rear wall extends to about 1 inch from the bottom wall.

[0031] In an example, the elongate front wall extends to about 1.5 inches from the bottom wall.

[0032] In an example, wherein the elongate front wall extends to about 1.85 inches from the bottom wall.

[0033] Other examples are shown and described herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] FIG. 1 is a front perspective view of a bottom cap for a foam insulating panel, according to an example;

[0035] FIG. 2 is a cross-sectional endwise elevation view of the bottom cap of FIG. 1;

[0036] FIG. 3 is a top plan view of the bottom cap of FIG. 1;

[0037] FIG. 4 is a front perspective view of the bottom cap of FIG. 1 receiving a foam insulating panel;

[0038] FIG. 5 is a cross-sectional endwise elevation view of an exterior insulating system for providing exterior insulation to a building;

[0039] FIG. 6 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to an alternative example;

[0040] FIG. 7 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to another alternative example;

[0041] FIG. 8 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to another alternative example;

[0042] FIG. 9 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to another alternative example;

[0043] FIG. 10 is a top plan view of the bottom cap of FIG. 9;

[0044] FIG. 11 is a front perspective view of the bottom cap of FIG. 9 receiving a foam insulating panel;

[0045] FIG. 12 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to another alternative example;

[0046] FIG. 13 is a top plan view of the bottom cap of FIG. 12;

[0047] FIG. 14 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to another alternative example;

[0048] FIG. 15 is a top plan view of the bottom cap of FIG. 14;

[0049] FIG. 16 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to an alternative example;

[0050] FIG. 17 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to an alternative example;

[0051] FIG. 18 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to an alternative example; and

[0052] FIG. 19 is cross-sectional endwise elevation view of a bottom cap for a foam insulating panel, according to an alternative example.

[0053] It will be appreciated that the various components disclosed herein are not necessarily shown in proportion in these figures.

[0054] Other aspects and examples will become apparent upon reading the following description.

DETAILED DESCRIPTION

[0055] FIG. 1 is a front perspective view of a bottom cap 5 for a foam insulating panel (the panel itself not shown in FIG. 1), according to an example. FIG. 2 is a cross-sectional endwise elevation view of bottom cap 5, and FIG. 3 is a top plan view of bottom cap 5.

[0056] In this example, bottom cap 5 includes an elongate rear wall 12 and an elongate front wall 14 spaced from and extending parallel to elongate rear wall 12. In this example, each of rear wall 12 and front wall 14 is planar and extends in a longitudinal direction between a first end 8 and a second

end 10. Bottom cap 5 also includes a bottom wall 16 extending between rear wall 12 and front wall 14. In this example, bottom wall 16 also planar and extends in the longitudinal direction between first end 8 and second end 10, such that rear wall 12, bottom wall 16, and front wall 14 together form an open-topped channel.

[0057] A dashed line 7 is included in FIG. 2 not to represent an actual component of bottom cap 5, but to aid with understanding relative angles of components of bottom cap 5. Dashed line 7 extends from rear wall 12 to front wall 14, and in the endwise view at right angles to both walls (since they are parallel to each other).

[0058] In this example, bottom wall 16 extends at an obtuse inside angle 13 (an angle greater than 90 degrees but less than 180 degrees) from rear wall 12 to front wall 14. Because of this, and because rear wall 12 and front wall 14 extend longitudinally in parallel to one another, bottom wall 16 and front wall 14 form an acute inside angle 15 (an angle greater than 0 but less than 90 degrees). The obtuse inside angle 13 is intended to be such that the channel has a sloped bottom when rear wall 12 is vertically-oriented (such as it is in FIG. 1 and FIG. 2), with the sloped bottom sloping downwards from rear wall 12 towards front wall 14. The sloped bottom is for the purpose of coaxing moisture that may enter the open-topped channel to flow away from rear wall 12 and towards front wall 14.

[0059] It will be appreciated that, due to potential variability in installation of bottom cap 5, bottom cap 5 itself may not be installed such that rear wall 12 is perfectly vertical. Thus, if the obtuse inside angle 13 is too small, a variance in installation that may cause rear wall 12 to be off-vertical by a few degrees counter-clockwise when bottom cap 5 is installed will, in turn, cause bottom wall 16 to either become level (i.e., horizontal) or to slope towards rear wall 12. If bottom wall 16 is level then moisture will not be coaxed towards front wall 14, and if bottom wall 16 is sloped towards rear wall 12, then moisture will be coaxed towards rear wall 12. For this reason, it may be useful to ensure obtuse inside angle 13 is greater than the expected installation variability, such that even if rear wall 12 is off-vertical by a few degrees counter-clockwise when bottom cap 5 is installed, bottom wall 16 will still slope towards front wall 14. Therefore, it may be useful for the obtuse inside angle to be at least about 95 degrees (i.e. a downward slope of 5 degrees from a vertical rear wall 12), providing for approximately a 4 degree variation counter-clockwise from rear wall 12 being vertical, while still providing a frontward-sloped bottom wall 16. It may alternatively be useful for the obtuse inside angle to be about 97 degree (i.e. a downward slope of 7 degrees from a vertical rear wall 12), providing for about a 6 degree variation counter-clockwise from rear wall 12 being vertical, while still providing a frontward-sloped bottom wall 16.

[0060] Furthermore, it will be appreciated that due to manufacturing variability, a given bottom cap 5 may have a slightly-different obtuse inside angle 13 after manufacturing than that of another bottom cap 5. It would be useful to ensure that manufacturing tolerances as well as installation tolerances be borne in mind when selecting the obtuse inside angle 13.

[0061] Still further, it will be appreciated that, depending on the malleability of the material used for bottom cap 5—such as if bottom cap 5 is made from sheet metal—an installer may find that during shipment or handling the

angles between rear wall 12, front wall 14, and bottom wall 16 have changed since manufacturing such that they need adjustment at the time of installation. It will also be appreciated that a given installer may decide to create a bottom cap 5 from sheet metal anew, onsite during installation. Such an installer may wish to use a jig to help ensure that the angles are suitable for coaxing any moisture that eventually gets inside of the channel of bottom cap 5 towards front wall 14.

[0062] While the above description about obtuse inside angle 13 explains the practical utility of having obtuse inside angle 13 greater than a certain amount, it will similarly be appreciated that providing an obtuse inside angle 13 that is too great may be impractical. For example, if the obtuse inside angle 13 is close to 180 degrees, then although the slope downwards towards front wall 14 will be assured this will, in turn likely cause bottom wall 16 to be impractically long and for front wall 14 to also be impractically tall to ensure it can aid, along with rear wall 12, with receiving a bottom of a foam building panel into the open-topped channel. Furthermore, the overall height of such a bottom cap 5 would be quite large if the obtuse inside angle 13 were very large, leading to possible installation problems and possible aesthetic problems. For these reasons, it may be useful to ensure obtuse inside angle 13 is greater than an amount such as one of the amounts set forth herein, while also being less than an amount. In particular, it may be useful for the obtuse inside angle to be at least about 95 degrees (i.e. a downward slope of 5 degrees from a vertical rear wall 12) but no greater than 105 degrees (i.e. a downward slope of 15 degrees from a vertical rear wall 12). It may alternatively be useful for obtuse inside angle 13 to be at least about 97 degrees (i.e. a downward slope of 7 degrees from a vertical rear wall 12) but no greater than 102 degrees (i.e. a downward slope of 12 degrees from a vertical rear wall 12). It will be appreciated that other obtuse inside angles 13 may be implemented as desired or required for a given installation, that take into account the considerations set forth herein and other considerations as may arise in a manufacturing, shipping, or installation context.

[0063] With reference particularly to FIG. 2 and FIG. 3, the open-topped channel has a width W for receiving the bottom of a foam insulating panel. In this example, width W is about 4 inches, so as to receive and “cap” the bottom of a foam insulating panel that is about 4 inches thick. Furthermore, in this example, obtuse inside angle 13 is 97 degrees and acute inside angle 15 is accordingly 83 degrees. In this example, the rear wall 12, front wall 14 and bottom wall 16 are formed from a single piece of sheet metal, bent into position using a roll forming process or another process suitable for bending the single sheet to form walls 12, 14, 16. Because, in this example, the tops of both rear wall 12 and front wall 14 come to the same height when rear wall 12 and front wall 14 are both perfectly vertical, if rear wall 12 is 1 inch tall, then with the 4 inch width W, front wall 14 would be about 1.85 inches tall.

[0064] In this example, bottom cap 5 includes a first set of spaced pedestals 18 extending into the open-topped channel from a region of bottom wall 16 that is proximal to rear wall 12. The spaced pedestals 18 together supports a rear end of the bottom of a foam insulating panel when received within the open-topped channel so that the foam insulating panel does not rest directly on bottom wall 16. In this way, moisture that might come into the channel from the rear

while the foam insulating panel is received within the channel is generally able to flow along bottom wall 16, underneath the foam insulating panel and through the spaces between pedestals 18, towards front wall 14. Each one of pedestals 18 may be formed by a punch process that forms indentations into sheet metal from a bottom-facing side of bottom wall 16, thus forming pedestals that extend up into the open-topped channel. Pedestals 18 may be formed by other processes.

[0065] Furthermore, bottom cap 5 includes a set of spaced apertures 20 extending from inside the open-topped channel through a region of bottom wall 16 that is proximal to front wall 14, to the exterior of bottom cap 5. The spaced apertures 20 provide a point of drainage of any moisture within the channel that flows towards front wall 14, thereby to enable the moisture to escape from the channel. In this example, the spaced apertures 20 extend along the bottommost point in the sloped bottom wall 16, so that moisture reaching the bottommost point can escape.

[0066] In alternative examples, spaced apertures may instead extend along a point that is slightly farther up the slope, or even very slightly up front wall 14 from the bottommost point and may still drain substantial moisture before or after the moisture reaches the bottommost point. That is, if moisture builds up at the bottommost point it will back up and drain through spaced apertures even if they are not themselves coincident with the bottommost point. In alternative examples, multiple rows or an array of spaced apertures may be provided instead of or in addition to a row of spaced apertures 20, as long as the integrity of the overall structure is not unduly compromised by the number and frequency of the spaced apertures.

[0067] It will be appreciated that, if the width W between rear wall 12 and front wall 14 is similar to the thickness of the bottom of the foam insulating panel, then the foam insulating panel will not easily tend to seat flat against the sloped bottom wall 16 near to front wall 14. Therefore, moisture in the channel is generally able to flow uninhibited underneath the foam insulating panel, to reach the spaced apertures 20, and thereby escape the channel. Each one of apertures 20 may be formed by a punch process that forms apertures through bottom wall 16. Apertures 20 may be formed by other processes.

[0068] It will be appreciated that one or both of the set of pedestals 18 and the set of apertures 20 may, during manufacture/creation, be formed in a length of sheet metal prior to the sheet metal itself being formed into rear wall 12, front wall 14, and bottom wall 16. Alternatively, one or both of the set of pedestals 18 and the set of apertures 20 may, during manufacture/creation, be formed in a length of sheet metal after the sheet metal is formed into rear wall 12, front wall 14, and bottom wall 16. Alternative orders of such steps to ensure quality, ease of manufacture, and cost effectiveness may be used.

[0069] FIG. 4 is a front perspective view of bottom cap 5 receiving a foam insulating panel P having a thickness of about 4 inches, corresponding to a width W of about 4 inches. It can be seen that, when seated within bottom cap 5, foam insulating panel P rests on pedestals 18 (only one of which is visible in FIG. 4) and closely between both rear wall 12 and front wall 14, thus providing a generally wedge-shaped volume formed mainly by the bottom of foam insulating panel P, a portion of front wall 14, and bottom wall 16, in which moisture can flow generally freely under-

neath foam insulating panel P towards apertures 20 (only one of which is visible in FIG. 4) so that the moisture can escape to outside of the channel.

[0070] FIG. 5 is a cross-sectional endwise elevation view of an exterior insulating system 100 for providing exterior insulation to a building. System 100 is adapted for use in applying additional insulation to an existing building, but may be used while constructing buildings anew, as the builder sees fit. System 100 includes an insulating panel P, in this example formed from expanded polystyrene (EPS) and having a wall-facing side and an outward-facing side. Insulating panel P has a top and a bottom. At the top, which is adjacent to a soffit 90 of the building B and particularly a soffit channel supporting soffit 90 in its position, there is a slope S downward from the wall-facing side to the outward-facing side. In this example, slope S is a 45 degree bevel. In FIG. 1, only one insulating panel P is shown, though installations will typically involve several insulating panels P. Each insulating panel P is fastened to the wall with fasteners F.

[0071] System 100 also includes an auxiliary panel 40 connected between soffit 90 and insulating panel P, as described in more detail in the '817 patent to Ackerley. In FIG. 5, one auxiliary panel 40 is shown, though other installations may involve more than one auxiliary panel 40. Each auxiliary panel 40 is, in this example, perforated to permit the free flow of air through the auxiliary panel 40. In this manner, once installed in place, air may flow freely to and from within building via the perforated auxiliary panel 40 and past the sloped top and the soffit 90 itself. The sloped top is useful for preventing blockage of soffit 90 as compared to an unsloped top which would cover a segment of soffit 90 and therefore block the free flow of air. The sloped top also provides more insulation to the upper regions of the wall 120 of the building than would simply spacing a flat-topped insulating panel from the soffit 90. This is because the sloped top enables the insulating panel P to reach to the top of the wall thereby providing some improved coverage of the wall than would the absence of insulation.

[0072] While system 100 is useful with one or more insulating panels P with sloped tops because airflow is unblocked while still providing at least some additional insulation to the top of the wall, alternatives are contemplated in which installations involve simply using auxiliary panel 40 with standard insulating panels (those having non-sloped tops), where the standard insulating panels are spaced from the soffit 90 to provide airflow that meets the applicable building code(s).

[0073] Furthermore, alternatives are contemplated in which just the sloped-top insulating panel P is employed or provided, without provision or installation of auxiliary panel 40, in order to provide insulation without air blockage where the finishing provided by the auxiliary panel 40 is not necessarily required. Furthermore, a drainage or moisture management membrane may be placed over sheathing 122 to inhibit ingress of moisture into the sheathing 122. The wall 120 supports rafters R which project outwardly to a fascia 124.

[0074] As shown also in FIG. 5, system 100 includes bottom cap 5, which receives and thus "caps" the bottom of insulating panel P, while coaxing any moisture M that comes within the channel of bottom cap 5 frontwards to reach

apertures 20 and thus be conveyed out of the channel and away from the wall 120 along a graded ground surface G.

[0075] System 100 enhances the insulation of the wall 120, while providing-using bottom cap 5—both aesthetic and functional coverage/protection of the bottom of the insulating panel P that further enables moisture that may have found its way behind the coverage to escape from system 100.

[0076] FIG. 6 is cross-sectional endwise elevation view of a bottom cap 50A for a foam insulating panel, according to an alternative example. Bottom cap 50A is very similar to bottom cap 5, in that it also includes a rear wall 12A, a front wall 14A, and a bottom wall 16A, a set of pedestals 18A, and a set of apertures 20A, all in very similar relative position and orientation as those of bottom cap 5. However, bottom cap 50A provides an obtuse internal angle 13A that is less than obtuse internal angle 13, and an acute internal angle 15A that is accordingly greater than acute internal angle 15. That is, the frontwards slope of bottom wall 16A is less than that of bottom wall 16 of bottom cap 5. In this example, obtuse internal angle 13A is 95 degrees and acute internal angle 15A is 85 degrees. Because, in this example, the tops of both rear wall 12A and front wall 14A come to the same height when rear wall 12A and front wall 14A are both perfectly vertical, if rear wall 12A is 1 inch tall, then with the 4 inch width W, front wall 14A would be about 1.5 inches tall.

[0077] FIG. 7 is cross-sectional endwise elevation view of a bottom cap 50B for a foam insulating panel, according to an alternative example. Bottom cap 50B is very similar to bottom cap 5, in that it also includes a rear wall 12B, a front wall 14B, and a bottom wall 16B, a set of pedestals 18B, and a set of apertures 20B, all in very similar relative position and orientation as those of bottom cap 5. However, bottom cap 50B provides an obtuse internal angle 13B that is less than obtuse internal angle 13, and an acute internal angle 15B that is accordingly greater than acute internal angle 15. That is, the frontwards slope of bottom wall 16B is less than that of bottom wall 16 of bottom cap 5. In this example, obtuse internal angle 13B is 93 degrees and acute internal angle 15B is 87 degrees. Because, in this example, the tops of both rear wall 12A and front wall 14A come to the same height when rear wall 12A and front wall 14A are both perfectly vertical, if rear wall 12A is 1 inch tall, then with the 4 inch width W, front wall 14A would be in the range of about 1 and less than 1.5 inches tall.

[0078] FIG. 8 is cross-sectional endwise elevation view of a bottom cap 50C for a foam insulating panel, according to an alternative example. Bottom cap 50C is very similar to bottom cap 5, in that it also includes a rear wall 12C, a front wall 14C, and a bottom wall 16C, a set of pedestals 18C, and a set of apertures 20C, all in very similar relative position and orientation as those of bottom cap 5. However, bottom cap 50C provides an obtuse internal angle 13C that is greater than obtuse internal angle 13, and an acute internal angle 15C that is accordingly less than acute internal angle 15. That is, the frontwards slope of bottom wall 16C is greater than that of bottom wall 16 of bottom cap 5. In this example, obtuse internal angle 13C is 100 degrees and acute internal angle 15C is 80 degrees. Because, in this example, the tops of both rear wall 12C and front wall 14C come to the same height when rear wall 12C and front wall 14C are both

perfectly vertical, if rear wall 12C is 1 inch tall, then with the 4 inch width W, front wall 14C would be greater than 1.85 inches tall.

[0079] FIG. 9 is cross-sectional endwise elevation view of a bottom cap 50D for a foam insulating panel, according to an alternative example. Bottom cap 50D is very similar to bottom cap 5, in that it also includes a rear wall 12D, a front wall 14D, and a bottom wall 16D, a set of pedestals 18D (a “first” set of pedestals, in this example), and a set of apertures 20D, all in very similar relative position and orientation as those of bottom cap 5. Furthermore, bottom cap 50D provides an obtuse internal angle 13C that is the same as obtuse internal angle 13, and an acute internal angle 15C that is accordingly the same as acute internal angle 15. That is, the frontwards slope of bottom wall 16D is the same as that of bottom wall 16 of bottom cap 5. However, in this example, bottom cap 50D includes a second set of spaced pedestals 19D extending into the open-topped channel from a region of bottom wall 16D that is proximal to front wall 12D. In this example, each of the spaced pedestals 19D in the second set extends to a same height with respect to rear wall 12D as the spaced pedestals 18D in the first set. In particular, the bottom side of a foam insulating panel received within the open-topped channel of bottom cap 50D would rest generally horizontally upon the first and second sets of pedestals 18D, 19D. As such, each of pedestals 19D is itself taller than pedestals 18D, but extends from a region of bottom wall 16D that is lower in the slope.

[0080] FIG. 10 is a top plan view of bottom cap 50D, showing bottom cap 50D extending between a first end 8D and a second end 10D. As shown, the second set of pedestals 19D extends along a side of the set of apertures 20D that is opposite front wall 4D. While alternatives to this configuration are possible, it is generally useful to have apertures 20D as close to the lowest point of the sloped bottom wall 16D as can be managed, subject to the strength and integrity of the bottom cap being maintained. Pedestals 19D may be formed using a similar process to that of pedestals 18D or may, due primarily to their different heights, be formed and/or arranged with respect to other features of bottom cap 50D using a different process.

[0081] FIG. 11 is a front perspective view of bottom cap 50D receiving a foam insulating panel P having a thickness of about 4 inches, corresponding to a width W of about 4 inches. It can be seen that, when seated within bottom cap 50D, foam insulating panel P rests on pedestals 18D (only one of which is visible in FIG. 11) and also on pedestals 19D (only one of which is visible in FIG. 11), and closely between both rear wall 12D and front wall 14D, thus providing a generally wedge-shaped volume formed mainly by the bottom of foam insulating panel P, a portion of front wall 14D, and bottom wall 16D, in which moisture can flow generally freely underneath foam insulating panel P towards apertures 20D (only one of which is visible in FIG. 11) so that the moisture can escape to outside of the channel.

[0082] FIG. 12 is cross-sectional endwise elevation view of a bottom cap 50E for a foam insulating panel, according to an alternative example. Bottom cap 50E is very similar to bottom cap 5, in that it also includes a rear wall 12E, a front wall 14E, and a bottom wall 16E, and a set of apertures 20E, all in very similar relative position and orientation as those of bottom cap 5. In this example, however, rather than a set of spaced pedestals extending into the open-topped channel from a region of the bottom wall 16E that is proximal to the

rear wall 12E, a set of spaced ledges 21E extends into the open-topped channel from a region of the elongate rear wall 12E that is proximal to bottom wall 16E. The spaced ledges 21E together support a rear end of the bottom of a foam insulating panel when received within the open-topped channel so that the foam insulating panel does not rest directly on bottom wall 16E. In this way, moisture that might come into the channel from the rear while the foam insulating panel is received within the channel is generally able to flow along bottom wall 16E, underneath the foam insulating panel and through the spaces between ledges 21E, towards front wall 14E. Each one of ledges 21E may be formed by a punch process that forms indentations into sheet metal from a rear-facing side of rear wall 12E, thus forming ledges 21E that extend laterally into the open-topped channel. Ledges 21E may be formed by other processes.

[0083] Furthermore, bottom cap 50E provides an obtuse internal angle 13E that is the same as obtuse internal angle 13, and an acute internal angle 15E that is accordingly the same as acute internal angle 15. That is, the frontwards slope of bottom wall 16E is the same as that of bottom wall 16 of bottom cap 5.

[0084] FIG. 13 is a top plan view of bottom cap 50E, showing bottom cap 50E extending between a first end 8E and a second end 10E. As shown, the second set of pedestals 19E extends along a side of the set of apertures 20E that is opposite front wall 14E. While alternatives to this configuration are possible, it is generally useful to have apertures 20E as close to the lowest point of the sloped bottom wall 16E as can be managed, subject to the strength and integrity of the bottom cap being maintained.

[0085] FIG. 14 is cross-sectional endwise elevation view of a bottom cap 50F for a foam insulating panel, according to an alternative example. Bottom cap 50F is very similar to bottom cap 50E, in that it also includes a rear wall 12E, a front wall 14E, and a bottom wall 16E, a set of apertures 20E, and a set of spaced ledges 21F extending into the open-topped channel from a region of the elongate rear wall 12F that is proximal to bottom wall 16F, all in very similar relative position and orientation as those of bottom cap 50E. However, in this example, like bottom cap 50D, bottom cap 50F includes a set of spaced pedestals 19F extending into the open-topped channel from a region of bottom wall 16F that is proximal to front wall 12F. In this example, each of the spaced pedestals 19F in the set extends to a same height with respect to rear wall 12F as the tops of spaced ledges 21F. In particular, the bottom side of a foam insulating panel received within the open-topped channel of bottom cap 50F would rest generally horizontally upon the set of ledges 21F and the set of pedestals 19F.

[0086] The spaced ledges 21F and the pedestals 19F together support the bottom of a foam insulating panel when received within the open-topped channel so that the foam insulating panel does not rest directly on bottom wall 16F. In this way, moisture that might come into the channel from the rear while the foam insulating panel is received within the channel is generally able to flow along in inside facing surface of rear wall 12F, between ledges 21E, and along bottom wall 16F underneath the foam insulating panel and to reach the apertures 20F.

[0087] Furthermore, bottom cap 50E provides an obtuse internal angle 13E that is the same as obtuse internal angle 13, and an acute internal angle 15E that is accordingly the

same as acute internal angle 15. That is, the frontwards slope of bottom wall 16E is the same as that of bottom wall 16 of bottom cap 5.

[0088] FIG. 15 is a top plan view of bottom cap 50E, showing bottom cap 50E extending between a first end 8E and a second end 10E. As shown, the second set of pedestals 19E extends along a side of the set of apertures 20E that is opposite front wall 14E. While alternatives to this configuration are possible, it is generally useful to have apertures 20E as close to the lowest point of the sloped bottom wall 16E as can be managed, subject to the strength and integrity of the bottom cap being maintained.

[0089] Although examples have been described, those of skill in the art will appreciate that variations and modifications may be made without departing from the spirit, scope and purpose of the disclosure as defined by the appended claims.

[0090] For example, it will be appreciated that, in examples, the tops of front and rear walls do not necessarily need to reach the same heights when the front and rear walls are perfectly vertical. For example, a rear wall in an example may rise to a lesser height than the front wall, or may rise to a greater height than the front wall. However, it will be appreciated that there should be provided a region of the rear wall that sufficiently contains the rear of a foam insulating panel when the bottom of the panel—typically a horizontal or near-horizontal bottom—is seated inside the bottom cap while a region of the front wall sufficiently contains the front of the same foam insulating panel.

[0091] FIG. 16 is cross-sectional endwise elevation view of a bottom cap 50G for a foam insulating panel, according to an alternative example. Bottom cap 50G is very similar to bottom cap 5, in that it also includes a rear wall 12G, a front wall 14G, and a bottom wall 16G, a set of pedestals 18G, and a set of apertures 20G, all in very similar relative position and orientation as those of bottom cap 5. Similarly, bottom cap 50G provides an obtuse internal angle that is the same as obtuse internal angle 13, and an acute internal angle that is accordingly the same as acute internal angle 15. However, in this example, the tops of rear wall 12G and front wall 14G come to different heights when rear wall 12F and front wall 14F are both perfectly vertical. In particular, front wall 14G comes to a greater height than does rear wall 12G.

[0092] FIG. 17 is cross-sectional endwise elevation view of a bottom cap 50H for a foam insulating panel, according to an alternative example. Bottom cap 50H is very similar to bottom cap 5, in that it also includes a rear wall 12H, a front wall 14H, and a bottom wall 16H, a set of pedestals 18H, and a set of apertures 20H, all in very similar relative position and orientation as those of bottom cap 5. Similarly, bottom cap 50H provides an obtuse internal angle that the same as obtuse internal angle 13, and an acute internal angle that is accordingly the same as acute internal angle 15. However, in this example, the tops of rear wall 12H and front wall 14H come to different heights when rear wall 12H and front wall 14H are both perfectly vertical. In particular, front wall 14H comes to a lesser height than does rear wall 12H.

[0093] FIG. 18 is cross-sectional endwise elevation view of a bottom cap 50I for a foam insulating panel, according to an alternative example. Bottom cap 50I is very similar to bottom cap 5, in that it also includes a rear wall 12I, a front wall 14I, and a bottom wall 16I, a set of pedestals 18I, and a set of apertures 20I, all in very similar relative position and orientation as those of bottom cap 5. Similarly, bottom cap

50I provides an obtuse internal angle that the same as obtuse internal angle **13**, and an acute internal angle that is accordingly the same as acute internal angle **15**. However, in this example, the tops of rear wall **12I** and front wall **14I** come to different heights when rear wall **12I** and front wall **14I** are both perfectly vertical. In particular, front wall **14I** comes to a lesser height than does rear wall **12I**. Furthermore, each of rear wall **12I** and front wall **14I** have extensions down from bottom wall **16I**, which may be useful functionally or aesthetically.

[0094] FIG. **19** is cross-sectional endwise elevation view of a bottom cap **50J** for a foam insulating panel, according to an alternative example. Bottom cap **50J** is very similar to bottom cap **5**, in that it also includes a rear wall **12J**, a front wall **14J**, and a bottom wall **16J**, a set of pedestals **18J**, and a set of apertures **20J**, all in very similar relative position and orientation as those of bottom cap **5**. Similarly, bottom cap **50J** provides an obtuse internal angle that the same as obtuse internal angle **13**, and an acute internal angle that is accordingly the same as acute internal angle **15**. However, in this example, the tops of rear wall **12J** and front wall **14J** come to different heights when rear wall **12J** and front wall **14J** are both perfectly vertical. In particular, front wall **14J** comes to a lesser height than does rear wall **12J**. Furthermore, as compared with bottom cap **50I** of FIG. **19**, only front wall **14J** has an extension down from bottom wall **16J**, which may be useful functionally or aesthetically.

[0095] Various further combinations of aspects disclosed herein are possible as examples, such as the examples set forth in the combinations of multiple clauses below.

Clauses

[0096] Clause 1. A bottom cap for a foam building panel, the bottom cap comprising:

[0097] an elongate rear wall;

[0098] an elongate front wall spaced from and extending parallel to the elongate rear

[0099] a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel;

[0100] a first set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate rear wall; and

[0101] a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.

[0102] Clause 2. The bottom cap of clause 1, wherein the obtuse inside angle is from about 95 degrees to about 105 degrees.

[0103] Clause 3. The bottom cap of clause 2, wherein the obtuse inside angle is from about 97 degrees to about 102 degrees.

[0104] Clause 4. The bottom cap of clause 3, wherein the obtuse inside angle is 97 degrees.

[0105] Clause 5. The bottom cap of clause 3, wherein the obtuse inside angle is 102 degrees.

[0106] Clause 6. The bottom cap of clause 1, further comprising:

[0107] a second set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the second set extends to a same height with respect to the elongate rear wall as the spaced pedestals in the first set.

[0108] Clause 7. The bottom cap of clause 1, wherein the elongate front wall is spaced from the elongate rear wall by about 4 inches.

[0109] Clause 8. The bottom cap of clause 7, wherein the elongate rear wall extends to about 1 inch from the bottom wall.

[0110] Clause 9. The bottom cap of clause 8, wherein the elongate front wall extends to about 1.5 inches from the bottom wall.

[0111] Clause 10. The bottom cap of clause 8, wherein the elongate front wall extends to about 1.85 inches from the bottom wall.

[0112] Clause 11. A bottom cap for a foam building panel, the bottom cap comprising:

[0113] an elongate rear wall;

[0114] an elongate front wall spaced from and extending parallel to the elongate rear

[0115] a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel;

[0116] a set of spaced ledges extending into the open-topped channel from the elongate rear wall; and

[0117] a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.

[0118] Clause 12. The bottom cap of clause 11, wherein the obtuse inside angle is from about 95 degrees to about 105 degrees.

[0119] Clause 13. The bottom cap of clause 12, wherein the obtuse inside angle is from about 97 degrees to about 102 degrees.

[0120] Clause 14. The bottom cap of clause 13, wherein the obtuse inside angle is 97 degrees.

[0121] Clause 15. The bottom cap of clause 13, wherein the obtuse inside angle is 102 degrees.

[0122] Clause 16. The bottom cap of clause 11, further comprising:

[0123] a set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the second set extends to a same height with respect to the elongate rear wall as the spaced ledges.

[0124] Clause 17. The bottom cap of clause 11, wherein the elongate front wall is spaced from the elongate rear wall by about 4 inches.

[0125] Clause 18. The bottom cap of clause 17, wherein the elongate rear wall extends to about 1 inch from the bottom wall.

[0126] Clause 19. The bottom cap of clause 18, wherein the elongate front wall extends to about 1.5 inches from the bottom wall.

[0127] Clause 20. The bottom cap of clause 18, wherein the elongate front wall extends to about 1.85 inches from the bottom wall.

What is claimed is:

1. A bottom cap for a foam building panel, the bottom cap comprising:

an elongate rear wall;

an elongate front wall spaced from and extending parallel to the elongate rear wall;

a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel;

- a first set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate rear wall; and
- a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.
- 2. The bottom cap of claim 1, wherein the obtuse inside angle is from about 95 degrees to about 105 degrees.
- 3. The bottom cap of claim 2, wherein the obtuse inside angle is from about 97 degrees to about 102 degrees.
- 4. The bottom cap of claim 3, wherein the obtuse inside angle is 97 degrees.
- 5. The bottom cap of claim 3, wherein the obtuse inside angle is 102 degrees.
- 6. The bottom cap of claim 1, further comprising:
 - a second set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the second set extends to a same height with respect to the elongate rear wall as the spaced pedestals in the first set.
- 7. The bottom cap of claim 1, wherein the elongate front wall is spaced from the elongate rear wall by about 4 inches.
- 8. The bottom cap of claim 7, wherein the elongate rear wall extends to about 1 inch from the bottom wall.
- 9. The bottom cap of claim 8, wherein the elongate front wall extends to about 1.5 inches from the bottom wall.
- 10. The bottom cap of claim 8, wherein the elongate front wall extends to about 1.85 inches from the bottom wall.
- 11. A bottom cap for a foam building panel, the bottom cap comprising:
 - an elongate rear wall;
 - an elongate front wall spaced from and extending parallel to the elongate rear wall;

- a bottom wall extending at an obtuse inside angle from the elongate rear wall to the elongate front wall thereby to form an open-topped channel;
- a set of spaced ledges extending into the open-topped channel from the elongate rear wall; and
- a set of spaced apertures extending from inside the open-topped channel through a region of the bottom wall that is proximal to the elongate front wall.
- 12. The bottom cap of claim 11, wherein the obtuse inside angle is from about 95 degrees to about 105 degrees.
- 13. The bottom cap of claim 12, wherein the obtuse inside angle is from about 97 degrees to about 102 degrees.
- 14. The bottom cap of claim 13, wherein the obtuse inside angle is 97 degrees.
- 15. The bottom cap of claim 13, wherein the obtuse inside angle is 102 degrees.
- 16. The bottom cap of claim 11, further comprising:
 - a set of spaced pedestals extending into the open-topped channel from a region of the bottom wall that is proximal to the elongate front wall, wherein each of the spaced pedestals in the set of spaced pedestals extends to a same height with respect to the elongate rear wall as the spaced ledges.
- 17. The bottom cap of claim 11, wherein the elongate front wall is spaced from the elongate rear wall by about 4 inches.
- 18. The bottom cap of claim 17, wherein the elongate rear wall extends to about 1 inch from the bottom wall.
- 19. The bottom cap of claim 18, wherein the elongate front wall extends to about 1.5 inches from the bottom wall.
- 20. The bottom cap of claim 18, wherein the elongate front wall extends to about 1.85 inches from the bottom wall.

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